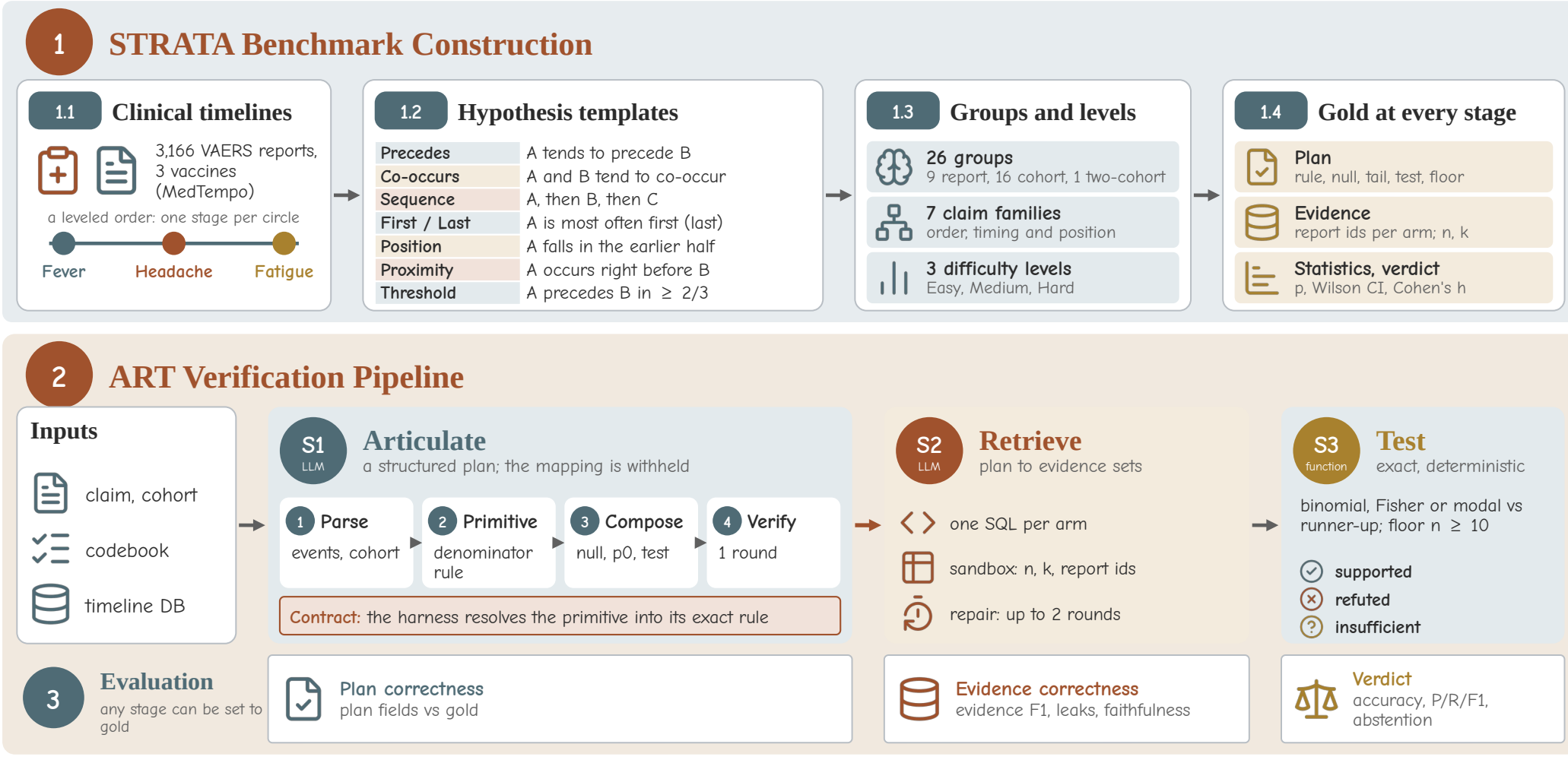

1
STRATA Benchmark Construction
1.1
Clinical timelines
3,166 VAERS reports, 3 vaccines (MedTempo)
a leveled order: one stage per circle
Fever
Headache
Fatigue
1.2
Hypothesis templates
Precedes
A tends to precede B
Co-occurs
A and B tend to co-occur
Sequence
A, then B, then C
First / Last
A is most often first (last)
Position
A falls in the earlier half
Proximity
A occurs right before B
Threshold
A precedes B in ≥ 2/3
1.3
Groups and levels
26 groups
9 report, 16 cohort, 1 two-cohort
7 claim families
order, timing and position
3 difficulty levels
Easy, Medium, Hard
1.4
Gold at every stage
Plan
rule, null, tail, test, floor
Evidence
report ids per arm; n, k
Statistics, verdict
p, Wilson CI, Cohen's h
2
ART Verification Pipeline
Inputs
claim, cohort
codebook
timeline DB
S1
LLM
Articulate
a structured plan; the mapping is withheld
1 Parse
events, cohort
2 Primitive
denominator rule
3 Compose
null, p0, test
4 Verify
1 round
Contract: the harness resolves the primitive into its exact rule
S2
LLM
Retrieve
plan to evidence sets
one SQL per arm
sandbox: n, k, report ids
repair: up to 2 rounds
S3
function
Test
exact, deterministic
binomial, Fisher or modal vs runner-up; floor n ≥ 10
supported
refuted
insufficient
3
Evaluation
any stage can be set to gold
Plan correctness
plan fields vs gold
Evidence correctness
evidence F1, leaks, faithfulness
Verdict
accuracy, P/R/F1, abstention